



VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, affiliated to Visvesvaraya Technological University, Belagavi

(Approved by AICTE, New Delhi & Government of Karnataka)

Accredited by **NBA (CV, CS, EE, EC, IS & ME)** | **NAAC with 'A' Grade**

P.B. No. 206, Gokulam III Stage, Mysuru-570 002, Karnataka, India

Phone: +91 821 4276201 /202 /225, Fax: +91 824 2510677

Web: <http://www.vvce.ac.in>

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Scheme of Teaching and Examination for BE (Autonomous Scheme: 2022)

VI SEMESTER - Computer Science & Engineering

Sl. No.	Course Area	Course Code	Course Name	Teaching Department	Contact Hours / week			Examination				Credits
					L	T	P	Duration (Hrs.)	CIE Marks	SEE Marks	Total	
1	IPCC	BCSML601	Machine Learning	CS	3	0	2	3	50	50	100	4
2	IPCC	BCSSS602	System Software and Compiler Design	CS	3	0	2	3	50	50	100	4
3	PC	BCSBT603	Blockchain Technology#	CS	3	0	0	3	50	50	100	3
5	PE	BCSXX614	Professional Elective - II	CS	3	0	0	3	50	50	100	3
6	OE	BCSXX655	Open Elective - II	CS	3	0	0	3	50	50	100	3
4	AEC	BCELK656X	Career Elective- II	CS	2	0	0	3	50	50	100	2
					0	0	4					
7	AEC	BITTP607	Technical Proficiency Enhancement Course-IV	CS	0	0	2	2	50	50	100	1
8	HS	BRIPK608	Research Methodology and Intellectual Property Rights	CS	1	0	0	2	50	50	100	1
9	AEC	BIKSK609	Indian Knowledge System	CS	1	0	0	2	50	50	100	1
TOTAL					17 19	0	6 10	-	450	450	900	22

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Professional Elective - II		Open Elective - II		Career Elective- II	
BCSFS614	Full Stack Development-II*	BCIBI655	Introduction to Business Intelligence	BCELK656X	Mini Project
BCSCC614	Cloud Computing*	BCSML655	Introduction to Machine Learning	BCELK656X	Research
BCSNS614	Network Security	BISWT655	Fundamentals of Cloud Computing	BCELK656X	Add on course
BCSDI614	Fundamentals of Digital Image Processing	BISFS655	Introduction to Full Stack Development	BCELK656X	Foreign language
		BISOS655	Introduction to Operating System	BCELK656X	Online Course

*common to (AI&ML/CSE/ISE), #common to (CSE/ISE), \$common to (AI&ML/CSE), @common to (AI&ML/ISE)



SEMESTER – VI

Course Name	: Machine Learning	Course Code :	BCSML601
No. of Lecture Hours / Week	: 03	CIE Marks :	50
No. of Tutorial / Practical Hours / Week	: 02	SEE Marks :	50
Total No. of Lecture + Tutorial / Practical Hours	: 40+24=64	SEE Duration :	03 Hrs.
L:T:P	: 3:0:2	Credits :	04
Course Prerequisites Basic Knowledge on variables, linear equations, graphs of functions, histograms, probability and statistics			
Course Overview This course is concerned with computer programs that automatically improve their performance through experience which illustrates methodologies, technologies, mathematics and algorithms of machine learning techniques.			
Course Learning Objectives (CLO) - This course will enable students to, - Define machine learning and understand the basic theory underlying machine learning. - Differentiate supervised, unsupervised and reinforcement learning - Understand the basic concepts of learning and decision trees. - Understand neural networks and Bayesian techniques for problems appear in machine learning - Understand the instant based learning and statistical analysis of machine learning techniques.			
Modules			Teaching Hours
Module 1 Introduction to Machine Learning Introduction, What is Human Learning? Types of Human Learning, What is Machine Learning, Types of Machine Learning, Problems not to be solved using Machine Learning, Application of Machine Learning, Issues in Machine Learning, Prepare to Model – Introduction, Machine Learning activities, Basic types of Data in Machine Learning SLT: Issues in Machine Learning Textbook1: Ch: 1.1 – 1.7, 1.9, 2.1 – 2.3			08
Supervised Learning: Learning a Class from Examples, Vapnik-Chervonenkis Dimension, Probably Approximately Correct Learning, Noise, Learning Multiple Classes, Regression, Model Selection and Generalization, Dimensions of a Supervised Machine Learning algorithm, Kernel Machines: Introduction, Optimal Separating Hyperplane, The Nonseparable Case: Soft Margin Hyperplane			08



SLT: v-SVM Textbook3: Ch: 2.1 -2.8, 13.1 – 13.4	
Module 3 Artificial Neural Networks Introduction, Neural Network representation, Perceptrons, Backpropagation algorithm, Remarks on the Backpropagation algorithm SLT: Appropriate problems Text Book2: Ch: 4.1 – 4.5	08
Module 4 Bayesian Learning Introduction, Bayes theorem, Bayes theorem and concept learning, ML and LS error hypothesis, MLH for predicting probabilities, MDL principle, Naive Bayes classifiers SLT: EM algorithm Textbook2: Ch: 6.1 – 6.6, 6.9, 6.11 - 6.11	08
Module 5 Instance Based Learning Introduction, k-nearest neighbor learning, locally weighted regression, Radial Basis Functions, Case-based reasoning, Reinforcement Learning: Introduction, The learning task, Q-Learning SLT: Remarks on Lazy and Eager Learning Textbook2: Ch: 8.1-8.6,13.1 – 13.3	08
<u>PRACTICAL MODULE-(PART-A)</u> 1. Implement and demonstrate the FIND-S Algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file 2. Write a program to demonstrate the working of SVM algorithm. Use an appropriate data set for building the model and apply this knowledge to classify new Samples. 3. Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets. 4. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets. 5. Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using the k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program. 6. Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Java/Python ML library classes can be used for this problem. 7. Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select the appropriate data set for your experiment and draw graphs. <u>Open Ended Enquiry Problems(PART-B)</u>	



1. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.
2. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Built-in Java classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.
3. Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set. You can use Java/Python ML library.

TYPE OF LAB EXERCISE PROBLEMS	LAB EXERCISE PROBLEM NUMBERS
DEMONSTRATION	1, 2
EXERCISE	3, 4
STRUCTURED ENQUIRY	5, 6, 7
OPEN ENDED	PART-B

Textbooks

1. Saikat Dutt, Subramanian Chandramouli, Amit Kumar Dos, "Machine Learning", 1st edition, Pearson, 2019
2. Tom M. Mitchell, Machine Learning, India Edition 2013, McGraw Hill Education
3. Ethem Alpaydın, Introduction to Machine Learning, 3rd Edition, The MIT Press

Reference Books

1. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, 2nd edition, springer series in statistics.
2. Ethem Alpaydın, Introduction to machine learning, second edition, MIT press
3. Foundations of Machine Learning, Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar, second edition, MIT Press, 2018.

Course Outcomes (COs)

At the end of the course students will be able to

CO1	Understand fundamentals of machine learning
CO2	Demonstrate the working of machine learning algorithms
CO3	Analyze input and optimization principles of machine learning for efficient and effective implementation
CO4	Design and implement machine learning techniques for solving complex problems in a team

CO - PO - PSO Matrix

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2											2		
CO2	3											2		
CO3		2										2		
CO4			2	2					2			2		
AVG	2.5	2	2	2					2			2		



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SEMESTER – VI

Course Name	: System Software and Compiler Design	Course Code:	BCSSS602
No. of Lecture Hours / Week	: 03	CIE Marks :	50
No. of Tutorial / Practical Hours / Week	: 02	SEE Marks :	50
Total No. of Lecture + Tutorial	: 40+24=64	SEE Duration :	03 Hrs.
L:T:P	: 3:0:2	Credits :	04
Course Prerequisites Basic Knowledge of Finite Automata and Computer Organization.			
Course Overview The course provides insights to the phases of a compiler. This course also aims to convey the language specifications, use of regular expressions, context free grammars and parsing techniques involved in the design of a compiler. The course introduces the basics of assemblers, loaders and linkers.			
Course Learning Objectives (CLO) This course will enable students to, - Understand the phases of compilers and lexical analyzer - Generate parse table, intermediate code, and target code. - Learn the concepts of System Software like assemblers and loaders.			
Modules			Teaching Hours
Module 1 Introduction: Language Processors, the structure of a compilers, Evolution of programming languages, the science of building a compiler Lexical Analysis: The role of lexical analyzer, Input Buffering, Specifications of Token, Recognition of Tokens. SLT: Applications of compilers technology Textbook 1: Ch. 1.1-1.5, 3.1 – 3.4			08
Module 2 Syntax Analysis - I: Introduction, Writing a grammar, Top-Down Parsers, Bottom-Up Parsers. SLT: Context Free Grammars Textbook 1: Ch. 4.1 - 4.5			08
Module 3 Syntax Analysis - II: Introduction to LR parsing: Simple LR Syntax Directed Translation: Syntax-Directed Definitions, Evaluation Orders for SDD's Intermediate Code Generation: Three-Address Code SLT: Variants of Syntax Trees Textbook 1: Ch. 4.6, 5.1-5.2,6.1 - 6.2			08
Module 4			08



Code Generation: The Target Language, basic blocks and flow graph, Optimization of basic blocks, The Code-Generation Algorithm Introduction to Assembler Machine Architecture of SIC and SIC/XE., Basic assembler functions SLT: Issues in the Design of a Code Generator Textbook 1: Ch. 8.1-8.2,8.4-8.5,8.6.2 Textbook 2: Ch. 1.3.1-1.3.2,2.1	
Module 5 Assembler: Machine dependent assembler features. Introduction to Loaders: Basic Loader Functions, Machine-Dependent loader features, Machine-Independent loader features SLT: Loader Design options Textbook 2: Ch. 2.2, 3.1-3.4	08
<u>PRACTICAL MODULE</u> 1) Write a LEX program to recognize valid arithmetic expressions. Identifiers in the expression could be only integers and operators could be + and *. Count the identifiers & operators present and print them separately. 2) Write a LEX program to eliminate comment lines in a C program and copy the resulting program into a separate file. 3) Write a LEX program to count the number of characters, words, spaces and lines in each input file. 4) Write a LEX program to recognize whether a given sentence is simple or compound. 5) Write YACC program to evaluate arithmetic expression involving operators: +, -, *, and /. 6) Write a YACC program to recognize valid identifier, operators and keywords in the given text (C program) file. 7) Write a YACC program to recognize a valid variable, which starts with a letter, followed by any number of letters or digits. 8) Develop, Implement and execute a program using YACC tool to recognize all strings ending with b preceded by n a's using the grammar aⁿ b (note: input n value). OPEN ENDED EXPERIMENTS	24



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Students can develop cross compilers, Bootstrap compilers, source to source compilers, decompilers or any other compiler applications in a team of maximum four members

Weightages:

Type of Experiment	Program Numbers	Weightage
Demonstration	1 2 & 3	33%
Exercise	4 5 & 6	33%
Structured Enquiry	7 & 8	24%
Open Ended		10%

Textbooks

1. Compilers-Principles, Techniques and Tools by Alfred V Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman. Pearson, 2nd edition, 2013
2. System Software by Leland. L. Beck, D Manjula, 3rd edition, 2016.

Reference Books

1. Systems programming – Srimanta Pal , Oxford university press, 2016
2. System programming and Compiler Design, K C Loudon, Cengage Learning
3. System software and operating system by D. M. Dhamdhare TMG
4. Compiler Design, K Muneeswaran, Oxford University Press 2013.

Course Outcomes (CO)

Upon the completion of the course, students will be able to

CO1	Explain the basic concepts of system software and compiler design
CO2	Apply the concepts of compiler design for parsing a given string and assembler features generating object program.
CO3	Analyze and trace the working of lexical analyser and syntax analyser for a given grammar
CO4*	Design and develop solutions to system software programs like LEX and YACC in a team

CO - PO - PSO Matrix

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3													
CO2	3											3		
CO3		2										2		
CO4			2		2			2						2
AVG	3	2	2		2			2				2.5		2



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SEMESTER – VI

Course Name	: Blockchain Technology	Course Code:	BCSBT603
Number of Lecture Hours / Week	: 03	CIE Marks:	50
Number of Tutorial / Practical Hours / Week	: 03	SEE Marks:	50
Total Number of Lecture + Tutorial/Practical Hours	: 40	SEE Duration:	03 Hrs.
L:T:P	: 3:0:0	Credits:	03
Course Prerequisites Basic knowledge of computer networks and programming skills			
Course Overview The course provides a platform for students to learn more about the history, the most important blockchain concepts, the philosophy of decentralization behind blockchain and the main discussions happening within the blockchain environment. In addition, applications of blockchain and the impact it could have on the business world.			
Course Learning Objectives (CLOs) - To understand basic concepts of blockchain. - To provide knowledge about blockchain by using new technologies. - To provide hands-on experience with the concepts.			
Modules			Teaching Hours
Module 1 Blockchain: Distributed systems, CAP theorem, Byzantine Generals problems, Consensus, Introduction to blockchain, various technical definitions of blockchains, Generic elements of a blockchain, Features of a blockchain, Applications of blockchain technology, Tiers of blockchain technology, Types of Blockchain ,Consensus in blockchain, CAP theorem and blockchain, Benefits and limitations of blockchain SLT: The history of blockchain Textbook: Ch1, Pg No.:10-32			08
Module 2 Decentralization: Decentralization using blockchain, Blockchain and full ecosystem decentralization, Smart contract, Decentralized organizations, Decentralized autonomous organizations, Decentralized autonomous corporations, Decentralized autonomous societies, Platforms for decentralization Cryptography and Technical Foundations: Mathematics, Cryptography, Cryptographic primitives: Symmetric cryptography, DES, AES SLT: Methods of decentralization Textbook: Ch2: Pg No.:34-37,40-48 & Ch3: Pg No.: 51-53,56-63			08
Module 3			08



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Cryptographic primitives: Asymmetric cryptography, Hash functions, Secure Hash Algorithms(SHAs),Merkle trees, Patricia trees, Distributed hash tables(DHTs), Digital signatures. SLT: Public and private keys-RSA Textbook: Ch3: Pg No.:65-78,87-98		
Module 4 Bitcoin: Bitcoin definition, Transactions, The transaction life cycle, The transaction structure, Blockchain: The structure of a block, The structure of a block header, Wallets. Smart Contracts: History, Definition, Ricardian contracts SLT: Types of transaction Textbook: Ch4: Pg No.:112-122,127-129,145-148 & Ch6		08
Module 5 Ethereum 101: Introduction, Ethereum clients and releases, Ethereum blockchain, Elements of the Ethereum blockchain, Ethereum virtual machine (EVM)- Execution Environment, Accounts, Block, Ether, Messages, Mining. SLT: The Ethereum stack Textbook: Ch7: Pg No.:210-227,235-238,244-254		08
Textbooks 1. Imran Bashir, "Mastering Blockchain", Packt, 2017		
Reference Books 1. Mastering Bitcoin: Programing the Open Blockchain Paperback-2017 by Andreas M. O'rielly 2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder. Bitcoin and cryptocurrency technologies: a comprehensive introduction. Princeton University Press, 2016.		

Course Outcomes (COs)

At the end of the course students will be able to

CO1	Explain the blockchain terminologies with its applications.
CO2	Illustrate tools and methods in Blockchain Technology.
CO3	Analyze the decentralized applications using Blockchain.
CO4	Apply ethical principles and demonstrate a private blockchain using Ethereum Tool.

CO - PO - PSO Matrix

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2													2
CO2	2													2
CO3		2												2
CO4					2			2						2
Avg.	2	2			2			2						2



SEMESTER – VI

Course Name	: Full Stack Development-II	Course Code:	BCSFS614
Number of Lecture Hours / Week	: 03	CIE Marks:	50
Number of Tutorial / Practical Hours / Week	: 00	SEE Marks:	50
Total Number of Lecture + Tutorial/Practical Hours	: 40	SEE Duration:	03 Hrs.
L:T:P	: 3:0:0	Credits:	03
Course Prerequisites Basic understanding of Programming skills and JavaScript.			
Course Overview The course provides a fundamental understanding of full-stack web development tools and techniques, with a particular focus on Node and ReactJS components of the MERN stack which a powerful combination of technologies that enables developers to build dynamic and scalable web applications			
Course Learning Objectives (CLOs) - Understanding of ReactJS runtime environment for JavaScript - Familiarize with ReactJS syntax and components for scripting - Acquaintance with ReactJS libraries for front-end web application development			
Modules			Teaching Hours
Module 1 What's in a Name? UI Layer, Virtual DOM, The Philosophy of React Thinking in Components Composition vs. Inheritance React Is Declarative React Is Idiomatic Why Learn React? React vs.... React vs. Angular 18 React vs. Vue What React Is Not React Is Not a Web Server React Is Not a Programming Language React Is Not a Database Server React Is Not a Development Environment React Is Not the Perfect Solution to Every Problem JSX Is Not HTML What Is JSX? How JSX Works Transpiler . . . Huh? Compilation vs. Transpilation JSX Transform Introducing Babel Eliminating Browser Incompatibilities Syntax Basics of JSX JSX Is JavaScript XML Beware of Reserved Words JSX Uses camelCase Preface Custom Attributes in DOM Elements with data- JSX Boolean Attributes Use Curly Braces to Include Literal JavaScript Remember to Use Double Curly Braces with Objects Put Comments in Curly Braces When to Use JavaScript in JSX Conditionals in JSX Conditional Rendering with if/else and Element Variables Conditional Rendering with the && Operator Conditional Rendering with the Conditional Operator SLT: Expressions in JSX Using Children in JSX React Fragments			08
Module 2 What Is a Component? Components vs. Elements Components Define Elements Elements Invoke Components Built-in Components HTML Element Components Attributes vs. Props Passing Props Accessing Props Standard HTML Attributes Non-Standard Attributes Custom Attributes User-Defined Components Types of Components Class Components Stepping			08



through a React Class Component React.Component Importing React.Component The Class Header The Constructor Function Managing State in Class Components The Render Function Creating and Using Props Function Components What Are Function Components? How to Write Function Components Optimizations and Function Component Shortcuts SLT: Managing State in Function Components Differences between Function and Class Components	
Module 3 One-Way Data Flow Understanding One-Way Data Flow Why One-Way Data Flow? Props Components Receive Props Props Can Be Any Data Type Props Are Read-Only Validating Incoming Props with PropTypes What Is PropTypes? Getting Started with PropTypes What Can PropTypes Validate? Default Props React State What Is state? Initializing state Initializing state in Class Components Initializing State in Function Components The Difference between state and props Updating state Updating a Class Component's state with setState Updating state with Function Components How Events Work in React What Is SyntheticEvent? Using Event Listener Attributes The Event Object Supported Events Event Handler Functions Writing Inline Event Handlers Writing Event Handlers in Function Components Writing Event Handlers in Class Components Binding Event Handler Functions SLT: Using bind Using Arrow Functions Passing Data to Event Handlers	08
Module 4 Forms Have State Controlled Inputs vs. Uncontrolled Inputs Updating a Controlled Input Controlling an Input in a Function Component Controlling an Input in a Class Component Lifting Up Input State Using Uncontrolled Inputs Using Different Form Elements Controlling the Input Element Controlling a textarea Controlling a Select Element Preventing Default Actions The Importance of Styles Importing CSS into the HTML File Using Plain Old CSS in Components Writing Inline Styles JavaScript Style Syntax Why to Use Inline Styles Why Not to Use Inline Styles Improving Inline Styles with Style Modules SLT: CSS Modules Naming CSS Module Files Advanced CSS Modules Functionality Global Classes Class Composition	08
Module 5 What Are Hooks? Why Were Hooks Introduced? Rules of Hooks The Built-in Hooks Managing State with useState Setting the Initial State Using the Setter Function Passing a Value to a Setter Passing a Function to a Setter Setter Function Value Comparison Hooking into the Lifecycle with useEffect Using the Default useEffect Behavior Cleaning Up After Effects Customizing useEffect Running Asynchronous Code with useEffect What Is Routing? How Routing Works in React Using React Router Installing and Importing react-router-dom The Router Component Selecting a Router Using the Router Component Linking to Routes Internal Linking with Link Internal Navigation with NavLink Automatic Linking with Redirect Creating Routes Restricting Path Matching Using URL Parameters The component Prop Render Props Switching Routes Rendering a Default Route Routing with Redirect Behind the Scenes: location, history, and match The history Object The location Object The match Object SLT: React Router Hooks useHistory useLocation useParams useRouteMatch	08



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Textbooks

1. Chris Minnick - Beginning ReactJS Foundations Building User Interfaces with ReactJS: An Approachable Guide, John Wiley & Sons, Inc. 2022

Reference Books

1. Basarat Ali Syed - Beginning Node.js-Apress, 2014.
2. Anthony Accomazzo, Ari Lerner, Clay Allsopp, David Guttman, Tyler Mcginnis, Nate Murray, FullStack React The Complete Guide to ReactJS & Friends, Fullstack.io, 2017

Course Outcomes (COs)

At the end of the course students will be able to

CO1	Explain various features of ReactJS for full stack web application development using MERN
CO2	Use various components and libraries of ReactJS to render view and perform corresponding operations
CO3	Examine application structure and their dependencies and interactions in an ReactJS application
CO4	Design front-end component of a web application using ReactJS library

CO - PO - PSO Matrix														
CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2											2		
CO2	3											3		
CO3		2										2		
CO4					2			2				2		
AVG	2.5	2			2			2				2.25		



SEMESTER – VI

Course Name : Cloud Computing **Course Code:** BCSCC614

No. of Lecture Hours / Week : 03 **CIE Marks:** 50

No. of Tutorial / Practical Hours / Week : 00 **SEE Marks:** 50

Total No. of Lecture + Tutorial : 40 **SEE Duration:** 03 Hrs.

L:T:P : 3:0:0 **Credits:** 03

Course Prerequisites

Basic Understanding of programming skills, familiarity with databases, basics of security and privacy, familiarity with operating systems and networking.

Course Overview

This course will emphasize on the on-demand delivery of compute power, database, storage, applications, and other IT resources through a cloud services platform via the internet. The course discusses the cloud infrastructure, application paradigms, resource management, security, google platforms, services and how to correctly use them in real-world applications.

Course Learning Objectives (CLO)

This course will enable students to,

- Identify the architecture, infrastructure and delivery models of cloud computing
- Compare and contrast different cloud services.
- Apply suitable virtualization concepts.
- Apply Cloud automation and management tools to build your own cloud application in Google Cloud Platform.

Modules	Teaching Hours
Module 1 Introduction to Cloud Infrastructure: Cloud computing, Cloud computing delivery models and services, Ethical issues, Cloud vulnerabilities, Major Challenges Faced by Cloud Computing, Cloudcomputing at Amazon, Cloud computing the Google perspective, Microsoft Windows Azure and online services, Open-source software platforms for private clouds, Cloud storage diversity and vendor lock-in, Energy use and ecological impact. SLT: Service level agreements, User experience and software licensing. Textbook 1: Ch. 1.3,-1.7, 3.1-3.8	08
Module 2 Cloud Computing: Application Paradigms and Concepts: Challenges of cloud computing, Architectural styles of cloud computing, Workflows: Coordination of multiple activities, Coordination based on a state machine model: The Zookeeper, The Map Reduce programming model.	08



SLT: The GrepTheWeb application. Textbook 1: Ch. 4.1-4.7		
Module 3 Cloud Resource Virtualization: Virtualization, Layering and virtualization, Virtual machinemonitors, Virtual Machines, Performance and Security Isolation, Full virtualization and Paravirtualization. Cloud Security- Cloud security risks, Privacy and privacy impact assessment, Trust, Operating system security. SLT: Virtual machine security. Textbook 1: Ch. 5.1- 5.6, 9.1-9.6		08
Module 4 Resource Management and Scheduling: Policies and mechanisms for resource management, Application of control theory to task scheduling on a cloud, Stability of a two-level resource allocation architecture, Feedback control based on dynamic thresholds, Coordination of specialized autonomic performance managers, A utility-based model for cloud-based Web services, Resourcing bundling: Combinatorial auctions for cloud resources, Scheduling algorithmsfor computing clouds. SLT: Fair queuing and Start-time fair queuing. Textbook 1: Ch. 6.1-6.10		08
Module 5 Google Cloud Platform and Services: Types of Cloud Services, Cloud Computing vs. Data Center Computing. Computing Components of Google Cloud Platform, Storage Components of Google Cloud Platform, Networking Components of Google Cloud Platform, Additional Components of Google Cloud Platform. How GCP Organizes Projects and Accounts, Roles and Identities. SLT: Billing and Enabling APIs. Textbook 2: Ch. 2, 3		08
Textbooks 1. Dan C Marinescu: Cloud Computing Theory and Practice, 2nd edition. Elsevier (MK) 2013. 2. Dan Sullivan: Official Google Cloud Certified Associate Cloud Engineer Study Guide, 1st edition, SYBEX, 2019		
Reference Books 1. John W Rittinghouse, James F Ransome: Cloud Computing Implementation, Management and Security, CRC Press 2013.		
Course Outcomes (CO) Upon the completion of the course, students will be able to		
C01	Explain the basic cloud computing concepts and distinguish between the various cloud infrastructures.	
C02	Illustrate different types of virtualizations, security and resource management techniques that can beused in designing cloud applications.	
C03	Perform analysis of the different virtualization and Resource Management techniques used in Google Cloud Platform.	
C04*	Investigate the various methods, algorithms for allocating and managing the resources in cloud.	



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CO – PO – PSO Matrix														
CO	PO											PSO		
	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02	PS03
C01	2											2		
C02	3											3		
C03		2										2		
C04					2			2				2		
AVG	2.5	2			2			2				2.25		



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SEMESTER – VI

Course Name	: Network Security	Course Code:	BCSNS614
No. of Lecture Hours / Week	: 03	CIE Marks :	50
No. of Tutorial / Practical Hours / Week	: 00	SEE Marks :	50
Total No. of Lecture + Tutorial	: 40	SEE Duration :	03 Hrs.
L:T:P	: 3:0:0	Credits :	03
Course Prerequisites Basic knowledge on computer networks and basic understanding of cryptography			
Course Overview This course aims to provide the concept of network security issues, services and goals, also to build protection mechanisms in order to secure computer networks. At the end of the course students will have the thorough knowledge on variety of network security applications and system-level security issues.			
Course Learning Objectives (CLO) This course will enable students to, • Explore knowledge on security concepts in terms of attacks, services, and mechanism • Analyze the security models of communication systems, networks and protocols used for secure transactions. • Understand various system security to protect against the threats in the networks.			
Modules			Teaching Hours
Module 1 Introduction to Security: Security Concepts, OSI Security Architecture, Security Attacks, Security services, Security Mechanism, A Model for Network Security SLT: Case studies on DoS Attack			08
Module 2 Authentication Applications: Kerberos – Motivation, Version 4, Version 5, X.509 Authentication Service, Public – Key Infrastructure SLT: Kerberos Encryption Techniques			08
Module 3 Email privacy: Pretty Good Privacy (PGP), S/MIME, IP Security: Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Combining Security Associations and Key Management. SLT: Case study on Replay Attack			08
Module 4 Web Security Web Security Consideration, Secure Socket Layer (SSL) and Transport Layer Security (TLS), Secure Electronic Transaction (SET).			08



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SLT: Threats on Web Security

Module 5

System Security

Intrusion Detection, Password Management, Distributed Denial of Service, Firewall-Design Principles

SLT: Honeypot & Types of Viruses

08

Textbooks

1. William Stallings, Cryptography and Network Security: Principles and Practice, PHI 4th Edition, 2006

Reference Books

1. Behrouz A. Foruzan, Cryptography and Network Security, Tata McGraw Hill 2007.
2. Cryptography and Network Security", by Atul Kahate 3rd edition Mc Graw Hill, 2008.

Course Outcomes (CO)

Upon the completion of the course, students will be able to

CO1	Understand the principles of Network security, System security and its issues
CO2	Apply the authentication application and services for security framework
CO3	Analyze Various security principles used for secure transactions
CO4*	Demonstrate different web and system security applications using modern tools

CO - PO - PSO Matrix

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2											2		
CO2	3											3		
CO3		2										2		
CO4					2			2				2		
AVG	2.5	2			2			2				2.25		



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SEMESTER – VI

Course Name	: Fundamentals of Digital Image Processing	Course Code: BCSDI614
Number of Lecture Hours / Week	: 03	CIE Marks: 50
Number of Tutorial / Practical Hours / Week	: 00	SEE Marks: 50
Total Number of Lecture + Tutorial/Practical Hours	: 40	SEE Duration: 03 Hrs.
L:T:P	: 3:0:0	Credits: 03
Course Prerequisites Basic knowledge of programming, linear algebra, data structures and algorithms.		
Course Overview This course introduces concepts like fundamentals of image formation, enhancement of images, segmentation, compression methods and object recognition.		
Course Learning Objectives (CLOs) - To understand image processing techniques - To understand the concepts of enhancement, segmentation, compression - To get familiarize with recognition and object detection methods		
Modules		Teaching Hours
Module 1 Introduction: What is Digital Image Processing, The Origins of Digital Image Processing, Examples of Fields that Use Digital Image Processing, Fundamental Steps in Digital Image Processing, Components of an Image Processing System Digital Image Fundamentals: Light and the Electromagnetic Spectrum, Image Sensing and Acquisition, Image Sampling and Quantization, Some Basic Relationships Between Pixels SLT: Elements of visual perception Textbook: Ch 1, 2		08
Module 2 Intensity Transformations and Spatial Filtering: Some Basic Intensity Transformation Functions Histogram Processing, Fundamentals of Spatial Filtering, Smoothing (Lowpass) Spatial Filters Sharpening (High pass) Spatial Filters Filtering in the Frequency Domain: The Discrete Fourier Transform of One Variable, Extensions to Functions of Two Variables, Some Properties of the 2-D DFT and Inverse DFT, The Basics of Filtering in the Frequency Domain, Image Smoothing Using Lowpass Frequency Domain Filters, Image Sharpening Using High pass Filters SLT: Homomorphic filtering Textbook: Ch 3, 4		08
Module 3		08



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Image Restoration and Degradation: A Model of the Image Degradation/Restoration process, Noise Models, Restoration in the Presence of Noise Only Spatial Filtering, Periodic Noise Reduction Using Frequency Domain Filtering

Color Image Processing: Color Fundamentals, Color Models, Color Transformations

SLT: Geometric mean filtering

Textbook: Ch 5, 6

Module 4

Image Compression: Fundamentals, Huffman Coding, LZW Coding

Morphological Image Processing: Preliminaries, Erosion and Dilation, Opening and Closing, The Hit-or-Miss Transform, Some basic Morphological operations- thickening, thinning, convex hull, skeletons

SLT: Pruning

Textbook: Ch 8, 9

08

Module 5: Image Segmentation: Fundamentals, Point, Line, and Edge Detection, Thresholding, Segmentation by Region Growing and by Region Splitting and Merging

SLT: Super pixels

Textbook: Ch 10

08

Textbooks

1. Digital Image Processing- Rafael C Gonzalez and Richard E. Woods, PHI 4th Edition 2013.

Reference Books

1. Fundamentals of Digital Image Processing- A K. Jain, Pearson 2004.
2. Image Processing analysis and Machine vision with Mind Tap by Milan Sonka and Roger Boile, Cengage Publications, 2018.
3. S.Jayaraman, S. Esakkirajan, T. Veerakumar, "Digital Image Processing", TataMcGrawHill

Course Outcomes (COs)

At the end of the course students will be able to

CO1	Explain fundamental image processing techniques and pattern recognition required for computer vision.
CO2	Apply enhancement, segmentation methods on images in image processing
CO3	Analyze image segmentation and compression techniques used in image processing
CO4	Explore and implement image processing applications for a given scenario in individual or a team

CO - PO - PSO Matrix

CO	PO											PSO		
	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	PS01	PS02	PS03
CO1	2											2		
CO2	3											3		
CO3		2										2		
CO4					2			2				2		
AVG	2.5	2			2			2				2.25		



SEMESTER – VI

Course Name	: Introduction to Machine Learning	Course Code: BCSML655
No. of Lecture Hours / Week	: 03	CIE Marks : 50
No. of Tutorial / Practical Hours / Week	: 00	SEE Marks : 50
Total No. of Lecture + Tutorial	: 40	SEE Duration : 03 Hrs.
L:T:P	: 3:0:0	Credits : 03
Course Prerequisites Basic Knowledge on variables, linear equations, graphs of functions, histograms, Probability and statistics		
Course Overview This course is concerned with computer programs that automatically improve their performance through experience which illustrates methodologies, technologies, mathematics and algorithms of machine learning techniques.		
Course Learning Objectives (CLO) This course will enable students to, • Define machine learning and understand the basic theory underlying machine learning • Differentiate supervised, unsupervised and reinforcement learning • Understand the basic concepts of learning and decision trees • Understand neural networks and Bayesian techniques for problems appear in machine learning • Understand the instant based learning and statistical analysis of machine learning techniques		
Modules		Teaching Hours
Module 1 Introduction: Well posed learning problems, Designing a Learning system, Perspective and Issues in Machine Learning. Concept Learning: Concept learning task, Concept learning as search, Find-S algorithm, Version space, Candidate Elimination algorithm. SLT: Inductive Bias Textbook1: Ch1.1,1.2,1.3 Ch2 2.2,2.3,2.4,2.5,2.7		08
Module 2 Decision Tree Learning: Decision tree representation, Appropriate problems for decision tree learning, Basic decision tree learning algorithm, hypothesis space search in decision tree learning, Inductive bias in decision tree learning. SLT: Why Prefer Short Hypotheses? Textbook1: Ch3.1-3.6		08
Module 3 Artificial Neural Networks: Introduction, Neural Network representation, Perceptron, Backpropagation algorithm. SLT: Adding Momentum Textbook1: 4.1-4.5.2.1		08
Module 4		08



Bayesian Learning: Introduction, Bayes theorem, Bayes theorem and concept learning, ML and LS error hypothesis, ML for predicting probabilities, MDL principle

SLT: Naive Bayes classifier

Textbook1:6.1-6.6,6.9

Module 5

Instance Based Learning: Introduction, k-nearest neighbor learning, locally weighted regression, radial basis function. Reinforcement Learning: Introduction, The Learning Task, Q Learning.

SLT: Cased Based Reasoning

Textbook 1: 8.1-8.5,13.1 to 13.3

08

Textbooks

1. Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar, Foundations of Machine Learning, 2nd edition, MIT Press, 2018
2. Tom M. Mitchell, Machine Learning, McGraw Hill Education, 2013

Reference Books

1. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, 2nd edition, Springer Series in Statistics, 2009
2. Ethem Alpaydm, Introduction to machine learning, 2nd edition, MIT press,2010

Course Outcomes (CO)

Upon the completion of the course, students will be able to

CO1	Explain the concepts of machine learning techniques
CO2	Illustrate the working of various machine learning algorithms
CO3	Analyze theory of probability and the concepts of neural network, classifier for problems appearing in machine learning
CO4*	Design and develop machine learning models for predictive analytics

CO - PO - PSO Matrix

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2													
CO2	2													
CO3		2												
CO4			2		2			2	1					
AVG	2	2	2		2			2	1					



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SEMESTER – VI

Course Name	: Technical Proficiency Enhancement Course - IV	Course Code : BITTP607
Number of Lecture Hours / Week	: 00	CCE Marks : 50
Number of Tutorial / Practical Hours / Week	: 02	SEE Marks : 50
Total Number of Lecture + Tutorial/Practical Hours	: 30	SEE Duration : 2 Hrs.
L:T:P	: 0:0:2	Credits : 01

COURSE PREREQUISITES: Working knowledge of C Programming and Data Structure.

COURSE OVERVIEW: This course is designed to enhance an individual's problem-solving, analytical, and critical thinking skills using competitive programming, graphs, strings and tries.

COURSE LEARNING OBJECTIVES (CLO) :

1. To enhance problem-solving skills.
2. To develop technical Skill in Recursion, Dynamic Programming, Graphs, Strings and Tries.
3. To Prepare students for job recruitment process and competitive exams.

MODULES	TEACHING HOURS
MODULE 1: Basics of Competitive Programming Introduction to Competitive Programming, Recursion, Time and Space Complexity Analysis, Language Tools, Searching and Sorting Applications, Two Pointers, and Sliding Window	06
MODULE 2: Recursion and Bit Manipulation Applications of Recursion: Advanced Recursion, Backtracking, Bit Manipulation and Modulo Arithmetic	06
MODULE 3: Dynamic Programming Dynamic Programming, Greedy Problems, Dp and Bitmasking	06
MODULE 4: Range Queries and Graphs Range Queries: Segment Tree, Fenwick Tree Graphs: Graph Implementation, Graph Algorithms, Advanced Graphs.	06
MODULE 5: Strings and Tries String Algorithms and Tries	06

[illegible]



SEMESTER – VI

Course Name : Research Methodology and Intellectual Property Rights **Course Code:** BRIPK608

No. of Lecture Hours / Week : 01 **CIE Marks :** 50

No. of Tutorial / Practical Hours / Week : 00 **SEE Marks :** 50

Total No. of Lecture + Tutorial : 20 **SEE Duration :** 02 Hrs.

L:T:P : 1:0:0 **CREDITS :** 01

COURSE OVERVIEW:

The course develops the skill of interpreting results and preparation of research reports and highlights intellectual property rights. Also, the course emphasizes various techniques of research. It involves developing theoretical and conceptual frameworks and writing a review. It sheds light on sampling designs and methods of data collection.

COURSE LEARNING OBJECTIVES (CLO):

1. To give an overview of the research methodology and explain the technique of defining a research problem.
2. To explain various research designs, sampling designs and different methods of data collections.
3. To explain the art of interpretation and the art of writing research reports and research proposals.
4. To explain various forms of the intellectual property, its relevance and business impact in the changing global business environment.

MODULES

TEACHING HOURS

MODULE 1: Introduction to Research Methodology

Introduction, meaning of research, Objectives of research, Types of research, Research approaches, Research methods versus Methodology, Research process.

Defining the Research Problem: Research problem, Selecting the problem, Necessity of defining the problem, Technique involved in defining a problem.

Textbook 1: Ch.1, Ch.2.

Self-Study Component: Criteria of good research.

04



MODULE 2: Research Design Introduction, meaning of research design, Features of a good design, Important concepts relating to research design, Different research designs, Basic principles of experimental designs- the Principle of Replication, the Principle of Randomization & Principle of Local Control. Textbook 1: Ch.3. Textbook 2: Ch.3, Ch.7, Ch.8. Self-Study Component: Need for research design.	04
MODULE 3: Sampling Design & Measurement Techniques Sampling Design: Introduction, Steps in sampling design, Characteristics of a good sample design, Different types of sample designs. Measurement and Scaling Techniques: Introduction, Measurement scales, Sources of error in measurement, Tests of sound measurement, Technique of developing measurement tools. Textbook 1: Ch.4, Ch.5. Self-Study Component: Criteria of Selecting a Sampling Procedure.	04
MODULE 4: Data Collection & Interpretation and Report Writing Data Collection: Collection of primary data-Observation Method, Interview Method, Collection of Data through Questionnaires, Collection of Data through Schedules, Collection of Secondary Data. Interpretation and Report Writing: Introduction, Technique of interpretation, Different steps in writing report, Layout of the research report, types of research report, Mechanics of writing a research report. Textbook 1: Ch.6, Ch.14. Self-Study Component: Selection of appropriate method for data collection.	04
MODULE 5: Intellectual Property Rights Introduction, The concept, Intellectual property system in India, Patents act-1970, Trademark act-1999, Protection of intellectual property under TRIPS, Copyright and related rights, Geographical indications, Industrial designs, Patents. Textbook 3: Ch.1. Self-Study Component: Trademarks	04
OTHER ASSESSMENT TOOLS (OAT): Case studies on research methodology & intellectual property rights and presenting the information in a report.	
Textbooks: 1. Research Methodology: Methods and Techniques, C.R. Kothari, Gaurav Garg, New Age International(P)Limited, Second Edition, 2019. 2. Research Methodology a step-by step guide for beginners, Ranjit Kumar, SAGE Publications Ltd., Fourth Edition, 2014. 3. Study material on Intellectual Property Rights-Law and Practice, The Institute of Company Secretaries of India, 2015.	



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Reference Books:

4. Research Methods: the concise knowledge base, Trochim Atomic Dog Publishing, 2004.
5. Conducting Research Literature Reviews: From the Internet to Paper, Fink A, Sage Publications, 2010.

COURSE OUTCOMES (CO): Upon the completion of the course, students will be able to

C01	Describe the knowledge of research techniques for conducting research studies
C02	Apply the knowledge of research process to provide solutions for engineering problems.
C03	Analyze the techniques of research for solving the problems
C04	Investigate the problems using research-based methods, appropriate techniques, resources, and engineering tools to provide valid conclusions

CO – PO Matrix

CO	PO										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
C01	2										
C02	2										
C03		2				2					
C04				2	2	-					
AVG	2	2		2	2	2					



SEMESTER-VI	
Course Name: Indian Knowledge System	Course Code: BIKSK609
No. of Lecture hours / week: 01	CCE Marks:50
No. of Tutorial hours / week: 00	SEE Marks:50
Total No. of Lecture Hours + Tutorial hours: 15	SEE Duration: 2Hr
L: T: P: 1-0-0	Credits: 1
PRE-REQUISITE : NIL	
COURSE OVERVIEW “We Owe a lot to the ancient Indians, teaching us how to count. Without which most modern scientific discoveries would have been impossible”- ALBERT EINSTEIN. The NEP, 2020 recognizes this rich heritage of ancient and eternal Indian knowledge and thought as a guiding principle. The Indian Knowledge Systems comprise of Jnana, Vignan, and Jeevan Darshan that have evolved out of experience, observation, experimentation, and rigorous analysis. This tradition of validating and putting into practice has impacted our education, arts, administration, law, justice, health, manufacturing, and commerce.	
COURSE LEARNING OBJECTIVES (CLO) 1. To facilitate the students with the concepts of Indian traditional knowledge and to make them understand the Importance of roots of knowledge system. 2. To make the students understand the traditional knowledge and analyse it and apply it to their day-to-day life	
MODULE	TEACHING HOURS
MODULE -I Indian Knowledge Systems (IKS)- An Introduction • Indian Knowledge Systems- An Overview: Need of Indian Knowledge System, Importance of ancient knowledge, Defining Indian Knowledge System, The IKS Corpus – A Classification Framework, Caturdaśa (Chathurdasha) – Vidyāsthāna (14 Divisions of IKS), Caturdaśa – Vidyāsthāna and The Income Tax Act, 1961. • History of IKS, Typical presentation Style- Sūtras, Encryptions, • The Vedic Corpus: Yajna and Project Management, Introduction to Vedas, The Four Vedas, four divisions of each Veda • Father- son conversation in Taittiriya- Upanishad, Indian Philosophical Systems – Development and Unique Features	3



• Wisdom through the Ages: Panchatantra- A Treatise on Statecraft through Stories, Puranas- An Encyclopedia, Maha-puranas, Upa-puranas and Sthala Puranas, Issues of Interest in puranas Self-Study Component (SSC): Unique Aspects of IKS Text book- 1	
MODULE – II Foundation Concepts for Science and Technology Computational concepts in Astadhyayi , Number system in India- Historical Evidence, Decimal System, Katapayadi System, Pingala and the Binary System. Self-Study Component (SSC): Measurement of time, distance and weight Text book- 1	3
MODULE – III Science in IKS Mathematics: Ancient Indians-Tryst with Mathematics, Unique aspects of Indian Mathematics, Great Mathematicians and their contributions, Arithmetic, Geometry, Trigonometry, Algebra. Astronomy: Unique aspects of Indian Astronomy, Historical Development of Astronomy in India, Cakra-Yantra Self-Study Component (SSC): Binary mathematics and Combinatorial problems in Chandah-Sastra of Pingala Text book- 1	3
MODULE – IV Engineering and Technology in IKS The Indian Science and Technology heritage, mining and ore extraction, Metals and Metal working Technology, Iron and Steel in India, Irrigation systems and practices in south India, Irrigation and Water Management, Dyes and Painting Technology, Surgical Techniques, Ship Building. Self-Study Component (SSC): Status of Indigenous Science and Technology Text book- 1	3
MODULE – V	3



Traditional Knowledge in Professional domain: Town planning and architecture, Health, wellness and Psychology, Governance and public administration, Traditional Medicine, Agriculture in ancient India.

Self-Study Component (SSC): Case studies on town planning

Text Book – 1, 2 & 3

Text Books

1. Introduction to Indian Knowledge System- concepts and applications, B Mahadevan, Vinayak Rajat Bhat, Nagendra Pavana R N, 2022, PHI Learning Private Ltd, ISBN-978-93- 91818-21-0
2. Traditional Knowledge System in India, Amit Jha, 2009, Atlantic Publishers and Distributors (P) Ltd., ISBN-13: 978-8126912230
3. Indian Knowledge System- Unveiling Traditions, Perceptives and Naratives, Dr. Ansul Saluja, 2023, Book Rivers.

Reference Books

1. Knowledge Traditions and Practices of India, Kapil Kapoor, Avadesh Kumar Singh, Vol. 1, 2005, DK Print World (P) Ltd., ISBN 81-246-0334
2. Traditional Knowledge Systems of India and Sri Lanka, A. V. Balasubramanian and T. D. Nirmala Devi, (2006), A Paper presented at the COMPAS Asian Regional Sri Lanka Workshop on Traditional Knowledge Systems and their Current Relevance and Applications, 3-5 July 2006, Bangalore.
3. Chemistry in Ancient India (Kannada) (Pracheena Bharatadalli Rasayana Vijnana), Kaiwar Gopinath, 2012, Rashtrorothana Sahitya, Bangalore
4. The Educational Heritage of Ancient India- How an Ecosystem of learning was laid to waste, Sahana Singh, 2017, Notion press.

Suggested Web Links:

1. <https://www.youtube.com/watch?v=LZP1StpYEPM>
2. <http://nptel.ac.in/courses/121106003/>
3. https://unctad.org/system/files/official-document/ditcted10_en.pdf
4. http://nbaindia.org/uploaded/docs/traditionalknowledge_190707.pdf
5. https://unfoundation.org/what-we-do/issues/sustainable-developmentgoals/?gclid=EAIaIQobChMImp-ltb_p8gIVTeN3Ch27LAmPEAAAYASAAEgIm1vD_BwE

COURSE OUTCOMES: At the end of the course student will be able to

C01	Appreciate richness of the Indian Knowledge System and importance of its learning relating to the present-day context.
C02	Apply the concepts of Indian knowledge System in different engineering domains



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C03

Analyze and solve the present day engineering problems using appropriate techniques, resources, and engineering concepts from the ancient Indian Knowledge System.

CO-PO-PSO MAPPING

CO\PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
C01	2					2					2			
C02			3			3								
C03			1	1		1								
AVG	2		2	1		2					2			